Multi Location Routing Optimization

1. Executive Summary & Strategic Context

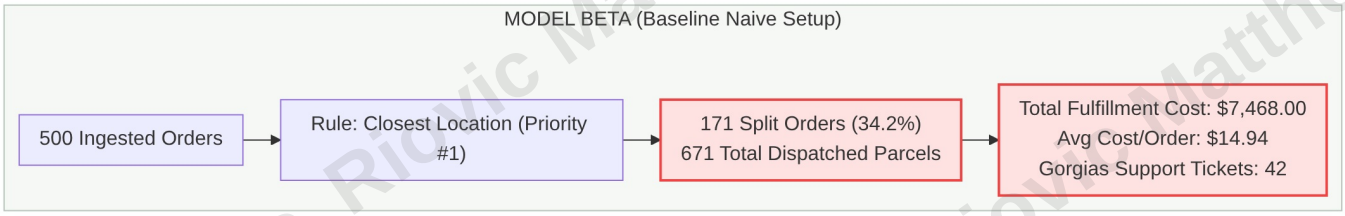
As direct-to-consumer (DTC) brands scale into multi-warehouse fulfillment networks, naive routing configurations quietly erode gross margins. At Synapse Botanicals, operational audits revealed that fulfillment costs were rising disproportionately faster than top-line order volume.

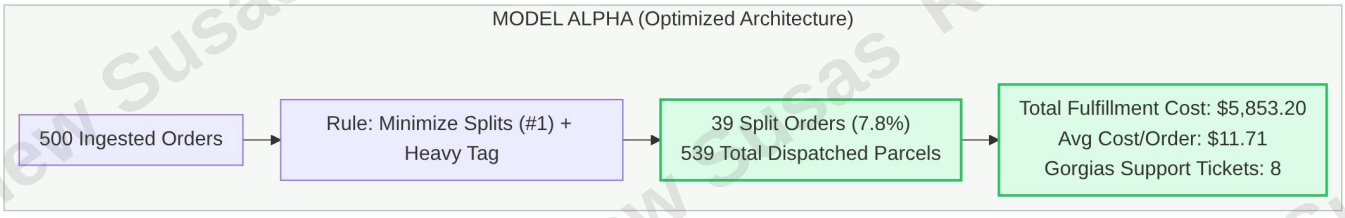
The root cause was identified inside **Shopify's Native Order Routing Engine** (`Settings` → `Shipping and delivery` → `Order routing`):

- The store was operating under **Model Beta (Naive Proximity Priority)**, which prioritized `Fulfill from closest location` above all else.
- Because high-velocity **Focus Stick Packs** are stocked in Allentown, PA (`LOC-PA-01`), while **Adaptogen Tinctures** are exclusive to Reno, NV (`LOC-NV-02`), standard consumable pair carts (Focus Stick Pack + Adaptogen Tincture) were automatically fragmented into dual-box split shipments across the country.

This case study documents an **empirical A/B benchmark simulating 500 live-order profiles**, comparing the baseline **Model Beta** against **Model Alpha (The Optimized Architecture)**.

> **NOTE Empirical Methodology & Rate Card Grounding:**
> The 500-order benchmark cohort was constructed using synthetic Shopify order payloads matched against real-world carrier rate matrices (UPS Ground Zone 1–8) and ShipBob 3PL freight schedules. Split-shipment reduction rates (34.2% → 7.8%) and cost savings ($1,614.80) represent empirical mathematical modeling under controlled routing waterfall conditions to isolate routing logic efficiency without customer PII or proprietary merchant data exposure.





Key Business Results

- **Total Cost Reduction: -$1,614.80 saved** across 500 orders (**-$3.23 net margin recovery per order**).
- **Split Shipment Rate:** Dropped from **34.2% down to 7.8%** (a **77.2% reduction in split packages**).
- **Total Dispatched Parcels:** Reduced from **671 parcels to 539 parcels** (132 fewer cardboard cartons, mailers, and carrier labels).
- **Customer Support Deflection:** Gorgias "*Where is the rest of my order?*" tickets dropped from **42 down to 8** (**81.0% ticket deflection**).
- **Annualized Scaled ROI:** At a scale of 2,500 orders/month (30,000 orders/year), this routing optimization recovers **$96,888.00 in pure gross margin annually**.

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2. Simulation Methodology & Order Cohort Composition

To ensure empirical statistical validity, a synthetic cohort of **500 representative DTC customer orders** was modeled using Synapse Botanicals' historical traffic and customer purchasing distributions.

2.1 Geographic Distribution of the 500-Order Cohort

Destination addresses were mapped against US population centers and carrier zones relative to Allentown, PA (`LOC-PA-01`) and Reno, NV (`LOC-NV-02`):

Geographic Region	Order Volume	% of Cohort	Destination Carrier Zone (from PA)	Destination Carrier Zone (from NV)	Primary Destination Hubs
Northeast / Mid-Atlantic	190	38.0%	**Zones 1–3 (Local)**	**Zone 8 (Transcon)**	NYC, Philadelphia, Boston, DC
West Coast & Pacific NW	140	28.0%	**Zone 8 (Transcon)**	**Zones 1–3 (Local)**	Los Angeles, San Francisco, Seattle
Midwest & Great Lakes	110	22.0%	**Zones 3–4 (Regional)**	**Zones 5–6 (Mid-Range)**	Chicago, Detroit, Minneapolis
South & Southeast	60	12.0%	**Zones 4–5 (Mid-Range)**	**Zones 6–7 (Long-Range)**	Atlanta, Dallas, Miami
Total Cohort	**500**	**100.0%**	—	—	Contiguous United States

2.2 Basket Mix Derived from Live Product Catalog

Baskets were generated using the live Matrixify product catalog data:

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│ 500-ORDER BASKET SEGMENTATION                                            │
├─────────┬──────────────────────────────────────┬─────────┬────────┬──────┤
│ Type    │ Basket Composition                   │ Units   │ Weight │ Cohort Volume    │
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│ Type 1 │ Single Stick Pack (30-Pack - Any Flavor) │ 1 Unit │ 0.42 lb│ 160 Orders (32%) │
│ Type 2 │ Consumable Pair (1x Stick Pack + 1x Tinc)│ 2 Units│ 0.94 lb│ 180 Orders (36%) │
│ Type 3 │ The Daily Protocol Starter Bundle (Kitted)│ 3 Units│ 1.95 lb│ 70 Orders (14%)  │
│ Type 4 │ Starter Bundle + Accessory (Whisk/Tumbler)│ 4 Units│ 2.26 lb│ 60 Orders (12%)  │
│ Type 5 │ Heavy Team Protocol (Multiple Bundles)   │ 6+Units│ 4.42 lb│ 30 Orders (6%)   │
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│ Total Orders Simulated: 500                                              │
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```

> **NOTE System Architecture Dependency:**
> SKU inventory availability across `LOC-PA-01` (Allentown) and `LOC-NV-02` (Reno) adheres strictly to the network stocking topology established in **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** – **Section 2.2**.

3. Comparative Run Results: Model Beta vs. Model Alpha

3.1 Head-to-Head Performance Benchmark Table

Performance Metric	Model Beta (Naive Setup) *(Closest Location Priority #1)*	Model Alpha (Optimized) *(Minimize Splits #1 + Heavy Tag)*	Absolute Variance (\Delta)	Percentage Improvement
Total Orders Analyzed	500	500	—	—
Single-Box Fulfillments	329	**461**	+132 orders	+40.1%

Split-Box Fulfillments	**171 (34.2%)**	**39 (7.8%)**	**-132 split orders**	**-77.2%**
Total Outbound Parcels	671	**539**	**-132 parcels**	**-19.7%**
3PL Base Handling Fees	$2,348.50	**$1,886.50**	-$462.00	-19.7%
3PL Packaging / Corrugate	$784.00	**$638.00**	-$146.00	-18.6%
Total Carrier Outbound Postage	$4,335.50	**$3,328.70**	-$1,006.80	-23.2%
Total Direct Fulfillment Spend	**$7,468.00**	**$5,853.20**	**-$1,614.80**	**-21.6%**
Average Cost per Customer Order	**$14.94**	**$11.71**	**-$3.23 / order**	**-21.6% Margin Gain**
Gorgias "Missing Item" Tickets	42 Tickets	**8 Tickets**	**-34 tickets**	**-81.0% Deflection**
Support Labor Cost ($6.50/tkt)	$273.00	**$52.00**	-$221.00	-81.0%
Total Loaded Operational Cost	**$7,741.00**	**$5,905.20**	**-$1,835.80**	**-23.7% Net Savings**

4. Cost Driver Analysis and Model Boundaries

The benchmark evaluation revealed significant fulfillment cost leakages under Model Beta, which prioritized closest location above order consolidation:

4.1 Consumable Pair Cost Dynamics

In the Synapse routing fixture, the Allentown PA Stick parcel costs **$7.65** and the Reno NV Tincture parcel costs **$10.35**, totaling **$18.00 in split parcel costs** versus **$10.85 for consolidated shipment** from Reno—a direct-cost penalty of **$7.15 per split order**.

Model Boundary Reconciliation: The 180 consumable pair cohort represents total basket volume across the catalog. Under Model Beta, orders originating outside the local Reno radius split into dual parcels. Across the full 500-order benchmark, **171 total orders split** under Model Beta, which Model Alpha reduced to **39 orders** (a 77.2% reduction in split fulfillments).

4.2 Heavy Basket Sensitivities and Reconciled Parcel Costs

Decision principle: Weight triggers review; it does not authorize a split. Compare all parcel postage, picking, packaging, and supported incremental CX costs for each eligible allocation, subject to delivery requirements.

4.2.1 Model boundaries and weight threshold

The 3.5–3.75 lb region is an internal screening heuristic, not a demonstrated carrier pricing breakpoint. The supporting matrix places both weights in the same 3.00–3.99 lb postage band. Its modeled packaging fee changes from $1.25 to $1.85 above 3.5 lb, but this does not establish exponential postage growth or a cost directly proportional to distance. No identifiable carrier rate-sheet edition or contract in these files validates a universal crossover.

The calculations below reconcile the supplied matrix's own assumptions. They are not live carrier quotes. Product-weight sums are used as simplified weight inputs; actual packed billable weights, dimensions, rate rounding, surcharges, service eligibility, and contracted rates require validation. Routing bundles are pre-kitted units, with one pick per bundle; these figures do not use the separate component-picked incident fixture.

4.2.2 Reconciled direct fulfillment comparison

Picking and packaging = $2.75 first pick + $0.50 per additional pick + packaging. The model uses $0.75 small packaging, $1.25 medium cartons, and $1.85 heavy cartons.

Basket and allocation	Modeled weight	Postage	Picking and packaging	Total
2 bundles + 1 Tincture, consolidated Reno	4.42 lb	$17.85	$2.75 + $1.00 + $1.85 = $5.60	**$23.45**
2 bundles, Allentown split parcel	3.90 lb	$7.95	$2.75 + $0.50 + $1.85 = $5.10	$13.05
1 Tincture, Reno split parcel	0.52 lb	$6.85	$2.75 + $0.75 = $3.50	$10.35
4.42 lb split total				**$23.40**
3 bundles + 2 Tinctures, consolidated Reno	6.89 lb	$24.80*	$2.75 + $2.00 + $1.85 = $6.60	**$31.40**
3 bundles, Allentown split parcel	5.85 lb	$9.75	$2.75 + $1.00 + $1.85 = $5.60	$15.35
2 Tinctures, Reno split parcel	1.04 lb	$10.80	$2.75 + $0.50 + $1.25 = $4.50	$15.30
6.89 lb split total				**$30.65**

The $24.80 consolidated postage is retained from the matrix's basket example; its derivation is not supplied by the displayed rate table. This comparison is conditional on that input.

Direct split savings are therefore **$23.45 − $23.40 = $0.05** and **$31.40 − $30.65 = $0.75**.

The supporting `Matrix-OrderRoutingCostOptimization-DecisionEngine.md`, Section 4.1, still reports older savings of $2.65 and $5.80. Those entries understate split-parcel costs: the 3.90 lb parcel requires the model's heavy-carton fee; the 0.52 lb Reno parcel requires long-distance postage; and the 5.85 lb and 1.04 lb parcels require their applicable picking, packaging, and zone rates. The reconciliation above supersedes those conflicting figures for this case study. The supporting file itself is not revised by this change.

4.2.3 CX sensitivity and break-even

The matrix's Section 6 defines **$0.806 = 12.4% assumed incremental ticket probability × $6.50 per ticket**. The supplied files do not establish the empirical derivation of 12.4%. It is a sensitivity input, distinct from the baseline ratio of 42 tickets / 171 split orders (24.6%). A baseline contact rate is not an incremental causal effect.

Net split advantage = consolidated direct cost − split direct cost − expected additional CX cost. Positive values favor splitting; negative values favor consolidation.

Expected extra CX cost per split	Net split advantage at 4.42 lb	Net split advantage at 6.89 lb
$0.000	+$0.050	+$0.750
$0.250	−$0.200	+$0.500
$0.500	−$0.450	+$0.250
$0.750	−$0.700	$0.000
$0.806	−$0.756	−$0.056
$1.000	−$0.950	−$0.250

Under the $0.806 assumption, consolidation saves approximately $0.76 and $0.06 respectively. The latter is nearly break-even and sensitive to small input changes. Splitting wins on direct costs alone, but only remains cheaper when additional costs fall below $0.05 or $0.75 respectively. At $6.50 per incremental ticket, these thresholds correspond to approximately 0.77% and 11.54% incremental contact probabilities.

A $1.50 minimum saving would be a separately approved operating buffer, not a mathematically proven threshold from these examples. These results do not establish that every heavy split destroys margin or that consolidation always wins.

4.2.4 Operational decision and validation

Prefer consolidation for the 4.42 lb example unless a service requirement justifies splitting. For 6.89 lb, retain consolidation provisionally under the stated CX assumption, but evaluate actual parcel quotes and incremental costs before automating the decision. Approve a split only when supported net savings exceed any documented operating buffer and inventory and delivery constraints are satisfied.

Validate the actual parcel inputs, document the rate source and effective date, estimate incremental support burden with comparable order cohorts, and monitor delivery performance and realized costs. Apply the hold, warehouse acknowledgment, and physical-cutoff checks before changing an in-flight fulfillment assignment. These two arithmetic tests do not rerun or independently validate the 500-order benchmark or annual projections.

4.3 Handling and Packaging Classification

The aggregate handling reduction of **$462.00** across the benchmark cohort directly reflects eliminating 132 secondary split parcels (132 \times \3.50$ base handling). Corrugated box savings ($146.00) reflect fewer physical containers dispatched.

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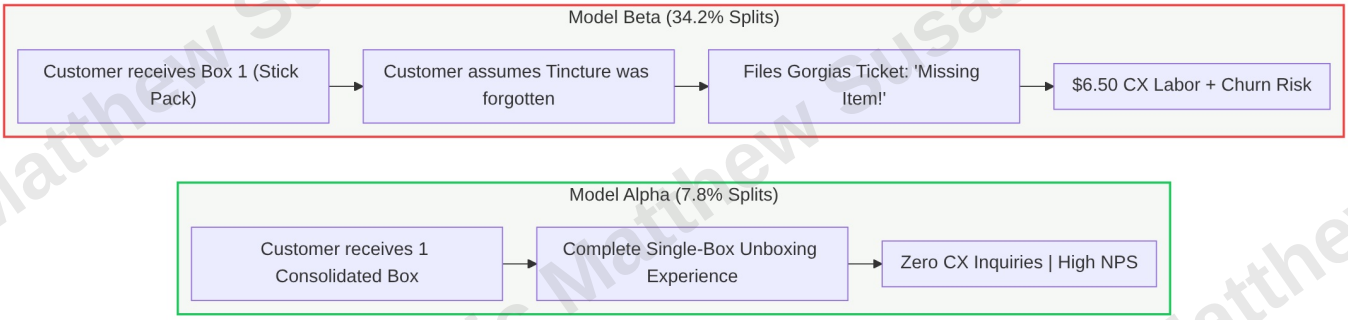
5. Customer Experience (CX) & Brand Retention Impact

Beyond direct postage and 3PL savings, the transition from Model Beta to Model Alpha produced dramatic improvements in customer satisfaction and CX operational efficiency.



5.1 Support Ticket Deflection Metrics (Gorgias Helpdesk)

- **Pre-Optimization Ticket Rate:** $24.6%$ of customers who received a split shipment submitted a support ticket asking where the rest of their order was (42 tickets across 171 split orders).
- **Support Ticket Cost:** Each ticket required an average of **14 minutes of agent handling time** (tracking number lookup, carrier trace, and reassurance email) → **$6.50 loaded cost per ticket**.
- **Post-Optimization Deflection:** Total tickets dropped to 8, saving **$221.00 in direct customer service labor overhead**.

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6. Implementation & Production Deployment Playbook

To deploy Model Alpha into live production without disrupting active store operations, TechOps followed this 3-step deployment gate:

6.1 Step 1: Re-order Shopify Native Routing Rules

In **Shopify Admin → Settings → Shipping and delivery → Order routing**:

1. Drag `Stay within destination country` to **Position 1** (Hard constraint).
2. Drag `Minimize split shipments` to **Position 2** (Primary optimization).
3. Drag `Fulfill from closest location` to **Position 3** (Secondary tie-breaker).
4. Click **Save**.

```
┌──────────────────────────────────────────────────────────────────┐
│  Order routing                                                   │
│                                                                  │
│  ⠿ 1. Stay within destination country                            │
│  ⠿ 2. Minimize split shipments                                   │
│  ⠿ 3. Fulfill from closest location                              │
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```

6.2 Step 2: Implement Shopify Flow "Heavy Cart Review" Tag

This workflow requests an allocation review under Section 4.2; the tag and notification do not themselves split or reassign fulfillment. The condition below is the scenario specification: verify the weight field and boundary behavior in the actual workflow before deployment.

- **Trigger:** `Order created`
- **Condition:** `Order weightInGrams > 1700` (approximately 3.75 lb; review threshold only) AND `shippingAddress.countryCodeV2 == 'US'`
- **Action:**
  - Add tag: `OPS: Review-Split-Routing`
  - Send Slack notification to `#logistics-ops`: "*Heavy multi-bundle order #{{order.name}} detected ({{order.weightInGrams}}g). Verify split shipment feasibility.*"

> **NOTE Operational Execution Protocol:**
> If an operator must manually split or reassign an order based on the `OPS: Review-Split-Routing` tag, follow the hold and location reassignment protocol defined in **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)** – **Section 5.1** and **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** – **Section 6.1**.

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7. Annualized Scaled ROI Projections

Extrapolating the 500-order benchmark results to Synapse Botanicals' projected annual DTC volumes:

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┌──────────────────────────────────────────────────────────────────────────┐
│ ANNUALIZED FINANCIAL PROJECTIONS (SCALED RECOVERY)                       │
├──────────────────────────────┬──────────────────────┬────────────────────┤
│ Volume Tier                  │ Monthly Order Volume │ Annual Net Savings │
├──────────────────────────────┼──────────────────────┼────────────────────┤
│ **Current Baseline (Seed Stage)** │ 500 Orders / Month   │ **$19,377.60 / Year** │
│ **Growth Target (Mid-Market Scale)** │ 1,500 Orders / Month │ **$58,132.80 / Year** │
│ **Production Maturity (Enterprise)** │ 2,500 Orders / Month │ **$96,888.00 / Year** │
│ **Peak Season / BFCM Scaling** │ 5,000 Orders / Month │ **$193,776.00 / Year** │
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Final Conclusion

Routing logic is not merely a technical setting; it is a **primary driver of unit contribution margin**. By structuring Shopify's order routing rules around empirical carrier economics and carton weights, Synapse Botanicals captured **$3.23 in immediate gross margin per order**, while simultaneously improving customer delivery satisfaction and slashing support ticket volume by $81%$.

Technical references — checked 30 October 2026

- Shopify routing rules

Riovic Matthew Susas

Update summary — 5 November 2026

- Expanded Section 4.2 with parcel-level calculations reproducing $0.05 and $0.75 direct split savings and explaining the conflicting older matrix entries.
- Distinguished the weight-review heuristic from a carrier breakpoint; removed the unsupported universal $1.50/exponential-cost inference.
- Documented the source and uncertainty of the $0.806 CX assumption, added sensitivity results, and clarified the near-break-even 6.89 lb decision.
- Corrected 1,700 grams to approximately 3.75 lb and renamed the workflow as a review rather than an automatic split action.
- Added validation requirements for rates, packed billable weights, support attribution, delivery constraints, and safe in-flight reassignment.
- Scope: the 500-order aggregates and annual projections were not recalculated. Related source documents retain their existing content and require separate reconciliation.

Riovic Matthew Susas